Xiong Ding

PHYSICS PH.D. · SOFTWARE ENGINEER

(+1) 678-882-9228 | dingxiong203@gmail.com | dingxiong.github.io/ | dingxiong | xiong-ding

Skills

Programming: C++, Python, Java
Skills: Kafka, MySQL, PostgreSQL, Elasticsearch, Kubernetes

Industry Experience

OpenAI Software Engineer San Francisco, CA, USA *04/2026 – Present*

Infrastructure

- Online Storage Team
  - Designed and implemented a peer-to-peer RPC topology for the file cache service, reducing Azure Blob Storage egress by 90%.
  - Optimized RocksDB for a latency-sensitive use case, reducing metadata cache misses by 95%.
  - Improved gRPC tail latency by approximately 50% through optimizations including multiple channels and Linux sendmsg with MSG_ZEROCOPY.
- **Tools & Technologies:** C++, RocksDB, Azure Blob Storage

Ziphq Software Engineer San Francisco, CA, USA *04/2022 – 04/2026*

Infrastructure

- Tech Lead, Core Infrastructure
  - Mentored junior engineers, helped them become domain owners, and guided them toward success.
  - Led quarterly team planning and took responsibility for priorities, execution, and outcomes.
  - Debugged the most challenging technical issues and production bugs.
- Launches & Major Projects
  - Migrated Zip from a single-data-center architecture to a multi-data-center architecture and implemented traffic routing strategies.
  - Migrated Ziphq's primary database from MySQL to Aurora PostgreSQL, improving performance by approximately 60% and reducing the storage footprint by approximately 50%.
  - Introduced Kafka; built the company-wide streaming platform and inbound webhook framework. Helped the payments team migrate workflows to the streaming platform.
  - Drove adoption of database CDC streams across multiple product teams.
  - Upgraded Kubernetes from v1.19 to v1.24.
  - Built numerous internal productivity tools for product engineering teams.
- **Tools & Technologies:** Python, Kafka, PostgreSQL, Kubernetes

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Citadel LLC Software Engineer Chicago, IL, USA *09/2019 – 03/2022*

Reconciliations and Control Infrastructure Team

- Reconciliation and Control System v3:
  - Co-designed the new reconciliation system from scratch; transferred domain knowledge from the legacy system to new team members.
  - Built core infrastructure components: Kafka C++ client, Datadog C++ client, logging modules, and more.
  - Developed the core matching service using Apache Arrow; optimized the hot path for maximum runtime performance.
- Other responsibilities:
  - Served as a primary contributor to disaster recovery (DR) planning and execution. Successfully executed multiple data center failovers with zero user impact.
  - Maintained and enhanced the v2 system; served as the point of contact for production issues.
  - Automated reconciliation workflows for futures, options, financial transmission rights (FTRs), and other OTC products.
- **Tools & Technologies:** C++, Boost, Apache Arrow, Spring Boot, Microsoft SQL Server, Kafka

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Airbnb Software Engineer San Francisco, CA, USA *05/2017 – 08/2019*

Search Ranking Team *03/2019 – 08/2019*

- Main contributions:
  - Migrated the heaviest batch job from the RDD API to the Dataset API, reducing runtime from 3 hours to 30 minutes.
  - Developed a new method to attribute user searches to final bookings, doubling the size of the training dataset.
- **Tools & Technologies:** Spark, Hive, Airflow, Scala

Home Infrastructure Team *05/2017 – 02/2019*

- Technical owner of the review platform:
  - Built the next-generation review service supporting multiple business verticals.
  - Migrated 340M+ reviews and 1B+ review categories to a new schema with zero downtime.
  - Partnered with the storage team to build a derived data store offering low-latency aggregated review metrics (counts, rating histograms, etc.), cutting tail latency by approximately 50%.
  - Migrated the EC2-hosted Elasticsearch cluster to AWS-managed Elasticsearch and upgraded the ES version from 1.4.5 to 6.2.1.
- **Tools & Technologies:** Dropwizard, Chef, Airflow, Elasticsearch, Mcrouter Cache, Powergrid (multithreading)

Education

Ph.D. in Physics	*GPA: 3.86/4.0*	Georgia Institute of Technology	Atlanta, GA, USA	*08/15/2012 – 05/05/2017*
M.S. in Computer Science & Engineering	*GPA: 3.86/4.0*	Georgia Institute of Technology	Atlanta, GA, USA	*01/01/2016 – 06/01/2016*
B.S. in Physics	*GPA: 3.75/4.0*	Wuhan University	Wuhan, China	*09/01/2008 - 06/05/2012*